1. Biological Background

1.1 Introduction

Animals exhibit an ability to physiologically compensate for the effects of environmental stress through phenotypic plasticity, either in development (developmental plasticity) or at maturity (reversible acclimation). For many species of fish, temperature and salinity represent two pervasive environmental stressors, where acclimation responses appear to be regulated through the thyroid axis (Little et al., 2013; Rejitha et al., 2009). In multi-stressor environments, however, overlapping roles of the axis may lead to signaling conflicts that undermine acclimation responses to one or both stressors. Specifically, both temperature and salinity represent energetic stressors: where temperature influences the thermodynamics of chemical reactions in ectotherms and salinity requires active transport for osmoregulation. Therefore, because of thyroid hormones (THs) conserved role in metabolic regulation across vertebrates, we predict there will be overlap in the metabolic responses during a prolonged acclimation to both conditions.

1.2 Thermal Acclimation

Ectotherms derive their bodily temperature from the environment, so their metabolic rate is subject to the thermodynamic effects of changes in external temperature. This is accomplished through shifts in metabolic enzymes and mitochondrial physiology where activity of enzymes including lactate dehydrogenase (LDH), citrate synthase (CS) and cytochrome C oxidase (COX) are adjusted to maintain sufficient energy balance throughout the thermal regime (Little et al., 2013; Seebacher et al., 2015). Specifically, enzyme activities are upregulated to enhance energy supplies during cold acclimation, and conversely metabolic suppression occurs in warm acclimation.

1.3 Salinity Acclimation

During movement between marine and freshwater habitats fish must rapidly acclimate to changes in salinity. This requires physiological processes that allow maintenance of osmotic equilibrium, as well as parallel regulation of metabolic activities to account for the cost of osmoregulation. Such a transition between physiologies is energetically expensive, placing high strain on the ATP reserves and metabolic state of the fish (Whitehead et al., 2012). This is compounded by the challenge of regulating ion flux for osmotic balance while simultaneously requiring ion movement coupling to mitochondrial electron transport-mediated ATP synthesis (Whitehead et al., 2012). Osmoregulation is performed primarily by mitochondria-rich ionocyte cells of the gill epithelium (Whitehead et al., 2012).

1.4 Experimental Design and Data Description

Mummichog killifish (*Fundulus heteroclitus*) were gradually acclimated to various combinations of either 11 parts per thousand (ppt) salinity or 28 ppt and 10 °C or 24 °C over three weeks. Then, gill tissue was sampled, homogenized and subject to a spectrophotometric assay which measures the maximal consumption of substrates for the enzymes LDH, CS and

COX. In the context of this study CS is used as a proxy for total (aerobic) mitochondrial content, LDH for anaerobic capacity and COX for (aerobic) mitochondrial capacity. Both CS and COX are used because changes aerobic capacity could be explained by mitochondrial biogenesis marked by CS and/or changes in mitochondrial structure marked by COX. Enzymatic activity assays were also conducted at both acclimation temperatures. Therefore, the experiment followed a trifactorial design (*Figure 1*) where, acclimation temperature, acclimation salinity and assay temperature can all be used as predictors. Therefore, each individual fish has two rows in the data file: one for each assay temperature. Additionally, body condition was measured to quantify if acclimation has measurable costs on the whole animal and was thereby chosen to be a response variable.

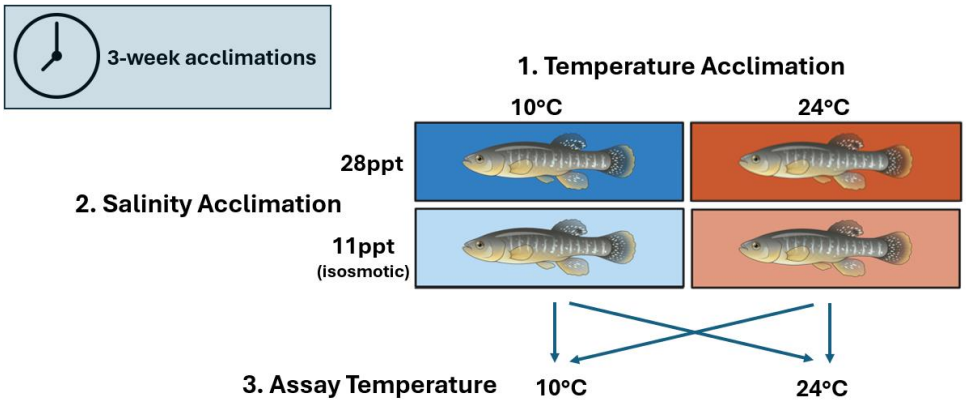

Figure 1. Representation of experimental design.

2. Aims and Hypotheses

Overall, this study aims to test whether overlapping metabolic demands of energetic stressors cause interactions that limit acclimation to either or both temperature and salinity. To accomplish this eurythermal and euryhaline mummichog killifish (*Fundulus heteroclitus*) were acclimated to four different temperature-salinity combinations (10 °C or 24 °C at 11 ppt or 28 ppt) to assess if these stressors interact in non-additive ways. Additionally, 11 ppt is isosmotic and therefore requires minimal ionoregulation so that the effects of temperature can be considered alone. Specific to this project, we seek to apply multivariate statistical models to the data set. This is because the response variables are not independent: both CS and COX are mitochondrial enzymes and a concurrent increase in LDH could be explained by an increase in total energetic capacity. Overall, we hypothesized that the combination of simultaneously acclimating to both temperature and salinity will undermine the responses characteristic of each stressor exposure in isolation. Following this, we predict that codependency between the metabolic responses during temperature and salinity acclimation drive an interaction between these predictors. Finally, we hypothesize that salinity acclimation (28 ppt) will act as a compounding stressor that buffers the thermal tolerance which will result in a lesser ability to acclimate to temperature.

3. Statistical Background

3.1 Redundancy Analysis (RDA)

Based on multivariate regression, RDA explains linear combinations of response variables based on linear combinations of the predictor variables (Capblancq & Forester, 2021). Using these linear combinations of the predictors, RDA maximizes the variance explained in the response variables (Capblancq & Forester, 2021). Therefore for this dataset, when an RDA is conducted with environmental predictors on the enzymatic activities, the constrained ordination axes, or axis of maximized variation, are comprised of covarying metabolic phenotypes that are correlated with the multivariate environment. We set out to use an RDA model because of its ability to model both multivariate predictors and multivariate responses. Specifically, we wanted to understand how predictors associated to the variation in responses across the data. As well, which responses covaried along the axes which maximized variation.

The required assumptions for an RDA model are that it identifies linear relationships among predictors and responses. We believe our data fits this assumption because we are interested in the potential differences in slopes across predictors and interactions between them. Using a polynomial RDA was considered because the Q10 effect of temperature is non-linear, but because each predictor is a factor with two levels, we believed a linear model was sufficient.

We predicted that, the enzyme activities would correlate together indicative of an upregulation of the metabolic phenotype. Within this, we also predicted that CS and COX activity would correlate the strongest since both are mitochondrial enzymes. As well, we predicted that within the RDA space, assay and acclimation temperature would load highly onto the primary RDA axes since temperature is likely a greater driver of metabolic demand than osmoregulation. Following this, we predict that individual data points will segregate by the predictors, where points at the same assay temperature will most highly associate with each other, followed by acclimation temperature then acclimation salinity.

3.2 Multivariate Analysis of Covariance (MANCOVA)

MANCOVA is an extension of a MANOVA which tests for the difference in vectors of means of multiple response variables between predictors (French et al., 2008). Additionally, it controls the effects of continuous covariates to remove confounding variance (). Therefore, a MANCOVA will be used to compare correlated changes in the enzymatic activity in response to the acclimation treatments. Similar to the RDA, a MANCOVA model was chosen because of underlying biological relation (dependency) between the enzyme functions, making a multivariate approach more impactful than considering all the responses independently.
The assumptions required for a MANCOVA model are like those for MANOVA, response variables should follow a normal distribution and homogeneity or equal levels of variance and covariance within response variables across the range of predictor variables. Additionally, MANCOVA requires absence of multicollinearity and homogeneity of regression slopes.

We sought to use MANCOVA to test the effect sizes and interactions between predictors. We predict that cold (10 °C) acclimation, high salinity (28 ppt) acclimation and warm assay temperature will increase the enzyme activities relative to within factor counterparts. Following this, we predict that assay temperature and acclimation temperature will have larger effect sizes

than acclimation salinity because temperature will place a greater metabolic demand than osmoregulation. We predict that an interaction will exist between the predictor's acclimation temperature and acclimation salinity because in conditions such as high salinity and warm temperature the metabolic profile should simultaneously be upregulated for osmoregulation and down regulated for temperature acclimation. We also predict an interaction between acclimation temperature and assay temperature because of temperature dependent isoforms/conformations of the enzymes that would be expressed during acclimation; where activity should be optimized when assay temperature matches the fish's acclimation temperature.

4. Results and Discussion

4.1 RDA

To model how metabolic phenotypes that are correlate with the multivariate environment, the predictors assay temperature, acclimation temperature and acclimation salinity were chosen and designated as factors. The responses of interest were continuous numerical values of enzymatic activity for COX, CS and LDH. Other responses, namely body condition and hematocrit were left out of the model so that focus could be placed on correlations between enzyme activities. These values were log transformed to place all the enzymatic activities on a similar scale; the untransformed values of LDH are approximately an order of magnitude greater than the other enzymes. Upon producing the model, diagnostic methods were used to ensure model fitness. A screeplot (*Figure 1A*) was used to determine which RDA axes would be used in further interpretation. RDA1 explains the majority of constrained variance, while the remainder is mostly encompassed in RDA2 (*Table 1*). Therefore, further analysis will focus on theses axes.

The majority of variance in the model is unconstrained and not associated with the predictors (*Table 2*), and the goodness-of-fit scores are low (*Figure 2B*). This means that most data points are represented poorly in the RDA space and there is variation unrelated to the predictors. As well, variation in the data does look homoscedastic as the range of residuals looks equally distributed across fitted values (*Figure 2C*). However, the residuals do not centre around 0 and are instead consistently >1. Therefore, this model appears to overall under predict the responses and suggests that predictors terms are missing from the model. Yet, the responses associate moderately with the RDA axes (*Table 3*) and the biplot scores of the predictors are very high with the RDA axes (*Table* 4). We believe this justifies that the RDA model can still have use in interpreting divergences in metabolic profiles associated with the predictors.

To test model significance permutation-based RDA was used. By randomly permutating the data and then comparing the observed results against this null distribution it allowed us to compare if there is a clear statistical relationship between our response variables and predictors. As well, we used a blocking factor (Fish_ID) because we had multiple measures for the same fish, one at each assay temperature, to define a restriction in the permutation. Our chosen threshold for significance was Pr >0.05. Overall, the relationship between the response matrix and predictor matrix was statistically clear (Pr = 0.005; *Table 5*), so we proceeded with permutation analysis by axis. Similar to our prediction that a single predictor (assay temperature)

would drive most of the variance, we see a statistically clear effect of RDA1 (Pr = 0.005; *Table 6*), and an unclear effect of RDA2 and RDA3 (*Table 6*). To then understand how the individual predictors contribute to that variance along each axis we performed a permutation analysis by term. Each of the three predictors were found to have a statistically clear effect on explaining the variance in the RDA environment (Pr = 0.005; *Table 7*). This told us that our predictor variables clearly explained the variation in the response variables and so we proceeded with interpretation of results using an RDA biplot.

RDA1 appears to be very strongly driven by assay temperature where cold temperature is on the left and warm on the right of the biplot (*Figure 3*). Similarly, RDA2 appears to be strongly driven by acclimation temperature where cold acclimation is to the top of the biplot and warm to the bottom (*Figure 3*). This result aligns with our prediction that temperature, both assay and acclimation, would be the strongest driver of metabolism causing them to be represented on RDA1 and RDA2 respectively. Acclimation salinity seems to sit between these axes, associating moderately with both axes, where higher salinity is in the top left and lower is in the bottom right. Following our predictions it appears that conditions that we predicted to increase metabolism, high salinity, higher assay temperature and cold acclimation, appear to trend towards the top left of the biplot space. Therefore, this region appears to be associated with factors that require metabolic demand and therefore data points that had relatively high enzyme activities. Contrary to our predictions, not all enzyme activities are increasing in this direction. COX activity is highly associated; however, CS and LDH appear to be correlating closely together towards the bottom left. This result is surprising because we assumed the greatest correlation would be between CS and COX. A difference between COX and LDH could be attributed to differences in the inherent aerobic versus anaerobic nature of the enzymes. Increased water temperature decreases the oxygen saturation capacity of water, placing greater reliance on anaerobic metabolism. In saltwater seabream, increasing temperature, or warm acclimation, increased LDH activity (Pimentel et al., 2019), which is contrary to decreases in LDH activity seen in freshwater species with warm acclimation (Little et al., 2013). This suggests that an interaction with salinity could exist, since saltwater also has reduced oxygen saturation compared to freshwater, where increases in LDH activity relative to aerobic metabolic enzymes is a compensatory mechanism for reduced oxygen availability in warm saltwater. However, this mechanism does not explain the disparity in CS activity from our predication where CS activity is in the same direction along RDA1, the assay temperature axis but trends in the opposite direction along RDA2 the temperature acclimation axis. In fact, the same study in seabream demonstrated an increase in CS activity with increasing temperature (Pimentel et al., 2019). Changes in CS and COX activity in response to temperature acclimation appears to species and tissue specific, for example warm acclimated lake whitefish have decreased CS activity in the liver and elevated CS and COX activity in the red muscle (Zak et al., 2017). Further modeling is clearly necessary to understand if this is a regulated response in *F. heteroclitus* or if an interaction between salinity and temperature acclimation is preventing the expected cold acclimation response.

Conclusively we believe the RDA model was an effective way to understand the overall metabolic response to each predictor alone, but also synchronously by using the biplot environment (*Figure* 3). It was especially useful for understanding the divergence between each enzyme activity which provided results contrary to our predictions. This demonstrated utility of using the RDA for monitoring differences in enzymes which were predicted to be highly correlated, CS and COX. We were also able to classify regions of the biplot to be associated with distinct metabolic characteristics, where the top left region appears to be associated with the predictors we thought would drive greatest metabolic demand, cold acclimation, warm assay temperature and osmoregulation. However, the model is clearly missing biological predictors and therefore, we decided to use body condition as a predictor going forward with MANOVA modelling.

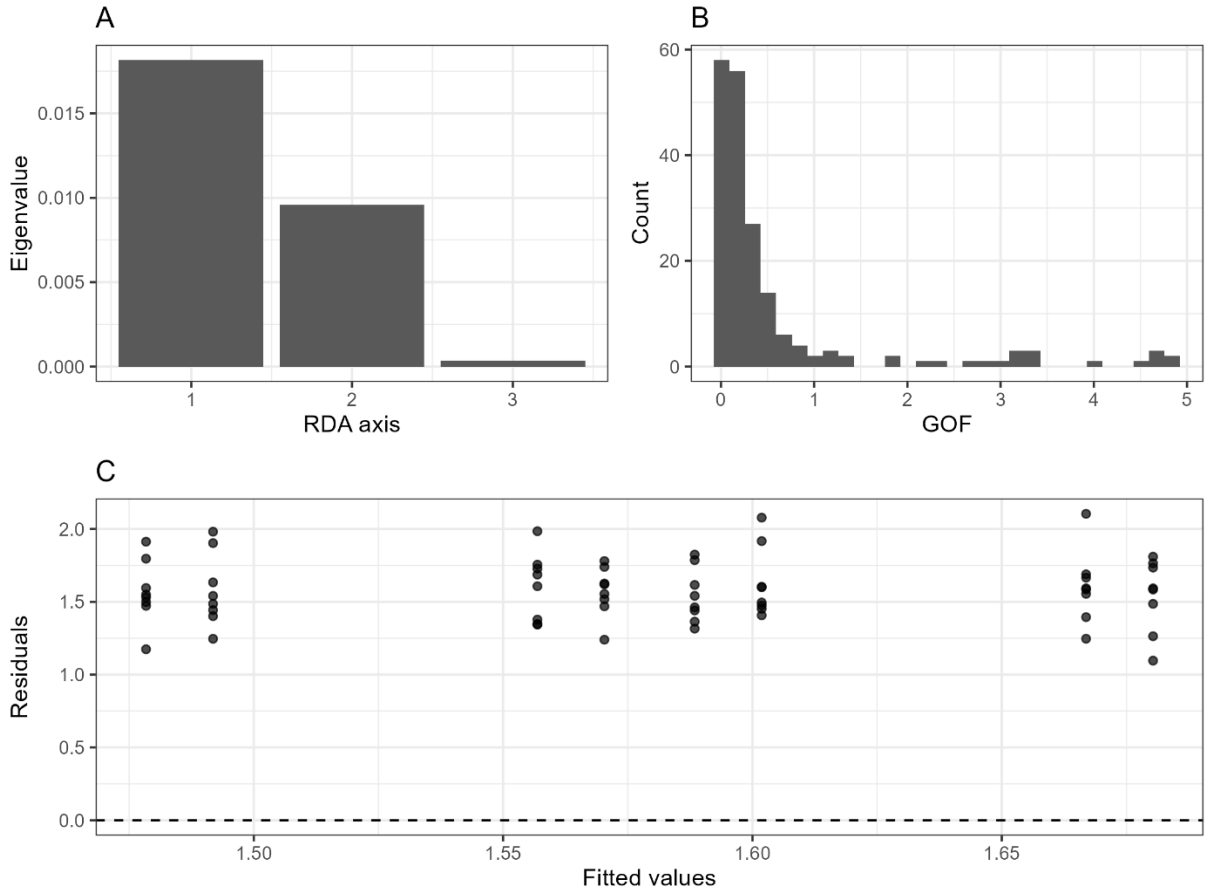

Figure 2. Diagnostic plots for the RDA model. A) Scree plot. **B)** Goodness-of-fit (GOF) plot. **C)** Fitted vs residuals plot.

Table 1. Eigenvalues of constrained axes of RDA.

	RDA1	RDA2	RDA3
Eigenvalue	0.018	0.009	0.000
Proportion Explained	0.646	0.341	0.013

Cumulative Proportion	0.646	0.987	1.000

Table 2. Partitioning of variance of RDA.

	Inertia	Proportion
Total	0.186	1.000
Constrained	0.028	0.151
Unconstrained	0.157	0.849

Table 3. Associations of responses with RDA axes.

	RDA1	RDA2
CS Activity	-0.228	-0.172
COX Activity	-0.323	0.344
LDH Activity	-0.422	-0.170

Table 4. Biplot scores of predictors with RDA axes.

	RDA1	RDA2
Assay Temperature (24)	-0.966	-0.174
Acclimation Temperature (24)	0.053	-0.854
Acclimation Salinity (28)	-0.252	0.489

Table 5. Permutation test for RDA under reduced model.

	Degrees of freedom	Variance	F	Pr (>F)
Model	3	0.028	3.571	0.005
Residual	60	0.157		

Table 6. Permutation test for RDA under reduced model by axis.

	Degrees of freedom	Variance	F	Pr (>F)
RDA1	1	0.018	6.920	0.005
RDA2	1	0.010	3.717	0.145
RDA3	1	0.000	0.141	0.480
Residual	60	0.157		

Table 7. Permutation test for RDA under reduced model by predictor term.

	Degrees of freedom	Variance	F	Pr (>F)
Assay Temperature	1	0.017	6.577	0.005
Acclimation Temperature	1	0.007	2.726	0.005
Acclimation Salinity	1	0.004	1.409	0.005
Residual	60	0.157		

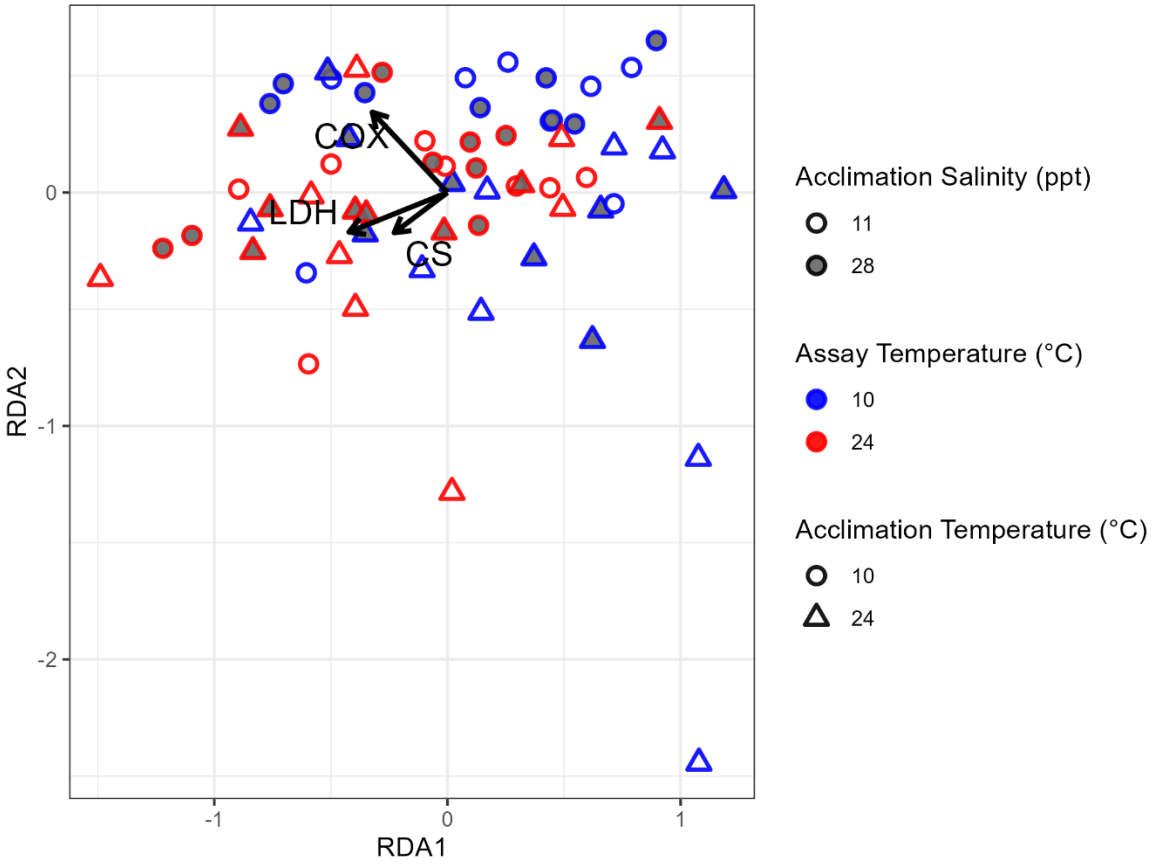

Figure 3. Redundancy analysis (RDA) biplot of axes RDA1 and RDA2.

4.2 MANCOVA

The initial model we suggested was a MANOVA, however, the RDA revealed that much of the variance was unaccounted for and the model was under predicting. Therefore, we sought to include body condition as a covariate because reduced metabolic activity could be explained by a decrease in the overall condition of the fish unrelated to plastic remodelling. The addition of a covariate had us pivot or model to a MANCOVA, however this model will still allow us to test for interactions between predictors. Following the RDA, enzyme activities were log transformed and categorical predictors were converted into factors.

Do determine if the assumptions were met, we first calculated he pairwise Pearson correlation between responses. We decided that a correlation coefficient (r) < 0.9 demonstrated an absence of multicollinearity and our responses met these criteria. Then to test homogeneity in the regression slopes, individual linear models were produced to test for statistically clear interactions between the covariate and responses. For this test we decided to set the threshold for violation at $p < 0.05$ and no statistically clear interactions were found between body condition and the enzyme activities. Then, a QQ-plot was produced which demonstrated the response variables follow a normal distribution (*Figure 4*). However, a Shapiro-Wilk multivariate normality test suggested that normality was not present ($W=0.781$, $p < 0.001$). We proceeded under the assumption that the QQ-plot demonstrated normality. Finally, a Box's homogeneity of covariance matrices test was also run to determine homogeneity of covariance. It is standard to reject the assumption of homogeneity if $p < 0.001$(Hahs-Vaugh, 2016). Our model did not meet this threshold ($p=0.002$) which demonstrated the data follow the assumption for homogeneity. Overall, we concluded our data met all the assumptions required for a MANCOVA analysis.

Proceeding with analysis we first tested for statistically clear interactions between our predictors, our chosen threshold for significance was $p < 0.05$. Following our predictions, we see that acclimation temperature and assay temperature alone had the only statistically clear effects on multivariate responses. From this we conclude that temperature is the greatest driving factor in metabolic responses and that even in a highly osmoregulatory tissue such as the gills, we do not appear to measure a clear metabolic cost of osmoregulation at the salinity used in the current experiment. Contrary to our predications, we do not detect and statistically clear two-way or three-way interactions between predictors.

To contextualize the magnitude of the effects, we looked at multivariate effect sizes. The cut off values chosen were 0.05 and 0.15 based on Cohen's d as there are not standard cut off values in the literature. 0.05 represents a mild effect and 0.15 represents a strong effect. As expected, assay temperature and acclimation temperature have the greatest effect on the enzyme activities *(Figure 5A).* Acclimation salinity and the interaction between assay temperature and acclimation temperature do show a mild effect although these predictors did not demonstrate a statistically clear effect ($p=0.19$ and $p=0.13$) *(Figure 5A).* Body condition which we thought would give insight to enzyme response, as the RDA underpredicted the effects, demonstrated a negligible influence on the activities *(Figure 5A)*. This means other biological factors need to be considered to explain the RDA results. The effect size of the two-way interaction between

acclimation salinity and acclimation temperature and the three-way interaction among acclimation salinity, acclimation temperature, and assay temperature appears to also be negligible. This was not expected and demonstrates that the predictors effect the enzyme response in an additive way not as an interaction. Biologically we believe this could be explained by the length of time of acclimation or that the energetic costs of acclimation were smaller than intended. The potential mild effect size suggests that some phenotypic remodelling is occurring. We believe that if the acclimation period was too short then the entirety of remodelling could be incomplete making it insufficient to determine if there was an interaction among the predictors. Alternatively, the thermal and salinity breadth used in the acclimation conditions are well within typical ranges for *F. heteroclitus* and the species is recognized for the eurythermal and euryhaline tolerance (Schulte, 2007). Thereby it could be that *F. heteroclitus* are especially adapted to multi-stressor environments making them robust enough to withstand these conditions without a breakdown in their acclimation response.

To determine the precision of these effect size values the 95% confidence intervals were inspected *(Figure 5B).* All confidence intervals had their upper bounds at 1. However, the lower bounds of the confidence intervals for assay temperature and acclimation temperature were not bound at 0. This demonstrated that only temperature and acclimation temperature have a clear effect on the enzyme activity compared to the other predictors.

To further investigate the influence each predictor had on the three enzyme activities, univariate responses were conducted with the significance threshold as $p<0.05$. COX activity was only influenced by acclimation temperature while LDH activity was only influenced by assay temperature. The predictors did not have a statistically clear influence on CS activity. To determine the extent these the predictors influence the enzyme activity, effect sizes were determined. Acclimation temperature had a mild effect on COX activity ($ETA^2= 0.08$) while acclimation temperature had great effect on LDH activity ($ETA^2= 0.19$). The effect the predictors have on CS activity were negligible. An interaction between temperature and salinity has been shown to influence CS activity, as well as an influence of these factors on their own (Chang et al., 2019). These results could similarly be explained by incomplete remodelling over too short of an acclimation period or minimal costs of acclimation. Regardless, univariate responses highlighted differences across enzymes. This showed us that although the intention of this project was to apply multivariate modelling to the data set, univariate analysis provides additionally context to our overall multivariate trends.

Overall, the MANCOVA demonstrated the predicted results that acclimation temperature and assay temperature would have the greatest influence on the enzyme activity. However, the interaction between acclimation temperature and acclimation salinity and the three-way interaction demonstrated to have minimal effect which was not predicted. The MANCOVA allowed us to determine the magnitude of effect the predictors had on the response variables to determine if there was a meaningful effect.

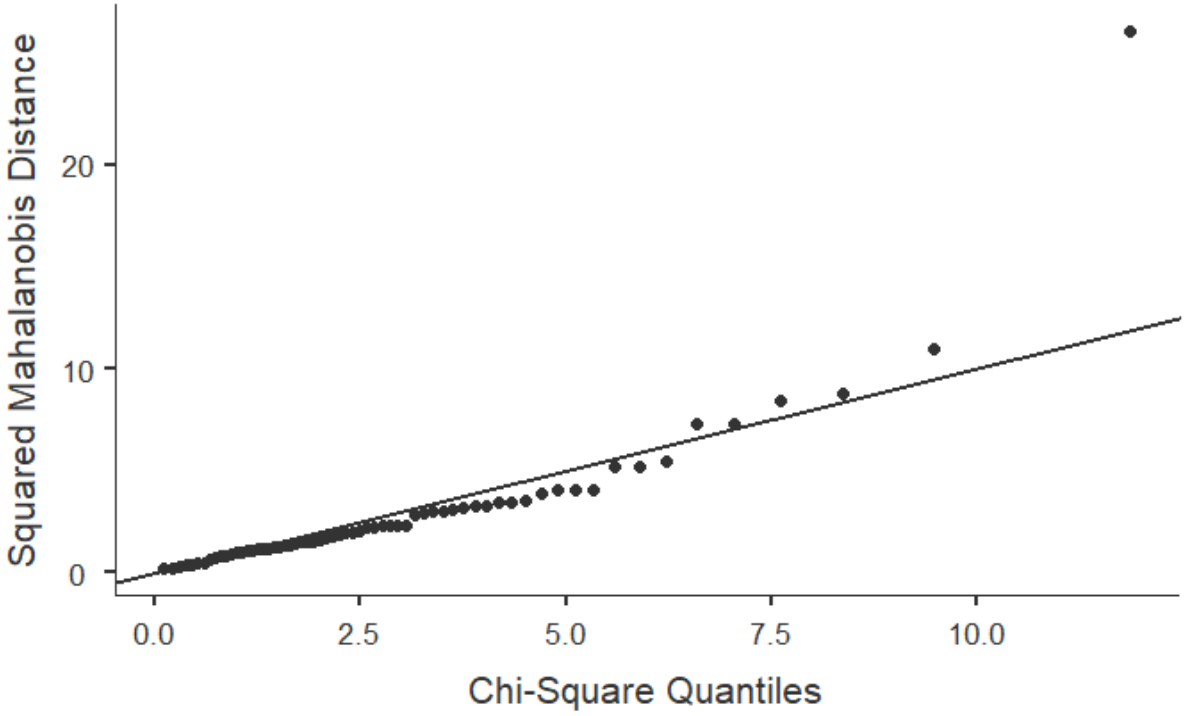

Figure 4. QQplot of MANCOVA model.

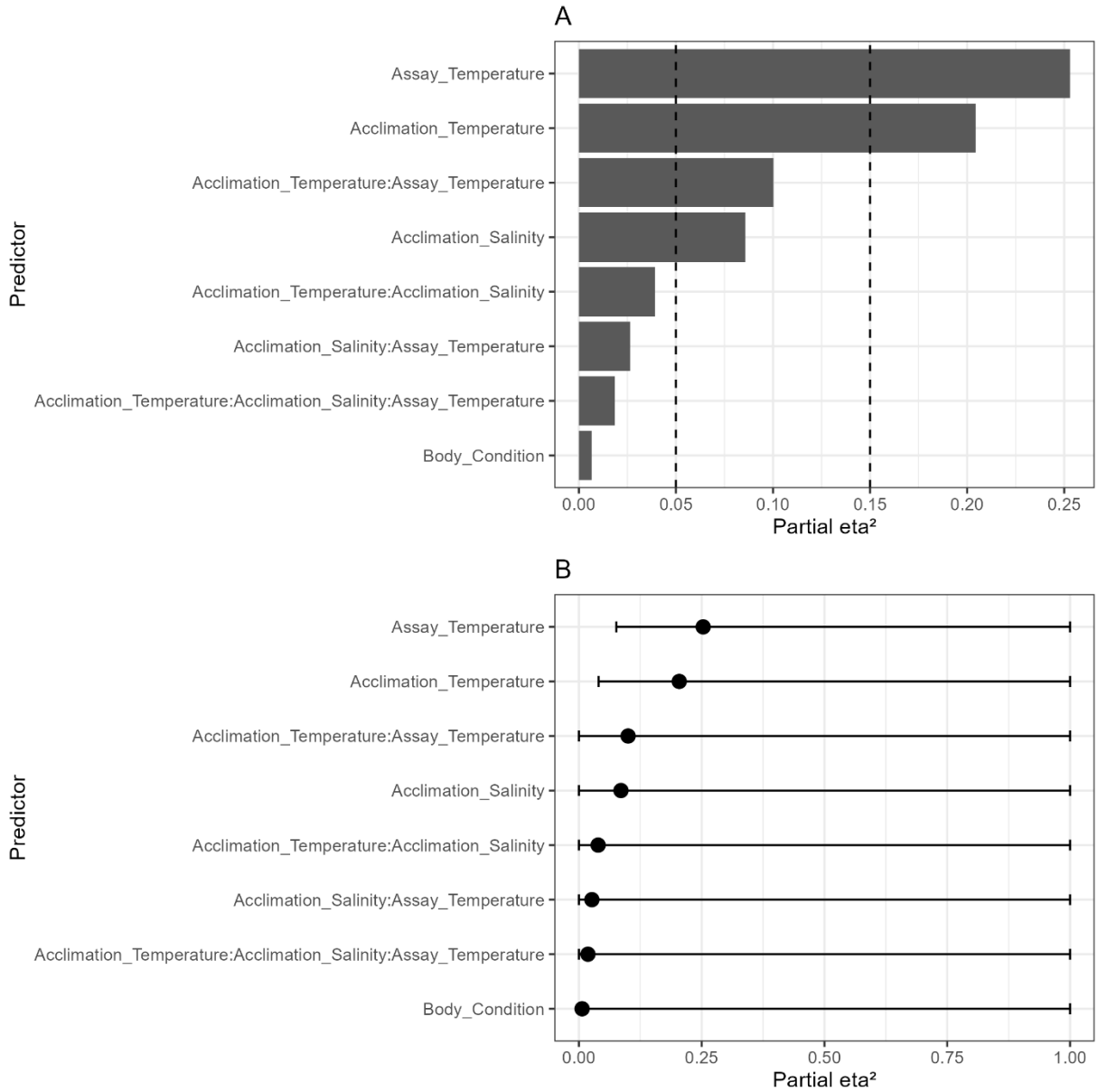

Figure 5. Effect size (partial eta^2) of MANCOVA model. A) Thresholds at 0.05 and 0.15. **B)** Confidence intervals with upper bounds fixed at 1.

5. Conclusion.

Environmental stressors such as temperature and salinity are obstacles that fish must overcome to survive. This study aimed to determine if the fish's ability to acclimate to the stressors, temperature and salinity, are hindered when exposed to both stressors at the same time. To determine this, killifish were acclimated at two temperatures and salinities for three weeks and then their mitochondrial enzymes COX, CS, and LDH were tested at the acclimation temperatures.

Multivariate analyses such as an RDA and MANCOVA were conducted to determine how the predictors acclimation temperature, acclimation salinity, and assay temperature impact the enzyme activities. The RDA demonstrated the expected results that temperature acclimation and assay temperature would have the greatest impact on the enzyme activities. However, the enzymes COX and CS did not correlated as predicted. The RDA allowed us to understand how the enzymes were affected by each predictor as well as seeing the whole interaction. However, the RDA was found to underpredict the responses as the residuals were not centred around zero. Therefore, another predictor must be used, and we decided to include body condition in our MANCOVA.

The MANCOVA was used to determine the predictors effect size for each response as well as if there was an interaction among the predictors. The results from the MANCOVA partially agreed with our predictions. Assay temperature and acclimation temperature had the greatest effect size with assay temperature having the greatest effect. The interaction between acclimation temperature and assay temperature was thought to influence the enzyme activities but only demonstrated a mild effect that was not statistically clear. As well, the expected interaction between acclimation temperature and acclimation salinity was not substantial and has a negligible effect on the enzyme activities. Finally, a lack of clear interactions demonstrated that assay temperature and acclimation temperature influence enzyme activities in an additive way rather than the expected interactive way.

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